6.1.3 Carboxylic acids and esters

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
Properties of carboxylic acids		
(a)	explanation of the water solubility of carboxylic acids in terms of hydrogen bonding	
(b)	reactions in aqueous conditions of carboxylic acids with metals and bases (including carbonates, metal oxides and alkalis)	Comparison of acidity of different carboxylic acids **not** required. **PAG7 (see 6.3.1 c)**

Esters

(c) esterification of:

- **(i)** carboxylic acids with alcohols in the presence of an acid catalyst (e.g. concentrated H_2SO_4)
- (ii) acid anhydrides with alcohols

(d) hydrolysis of esters:

- **(i)** in hot aqueous acid to form carboxylic acids and alcohols
- (ii) in hot aqueous alkali to form carboxylate salts and alcohols

Acyl chlorides

(e) the formation of acyl chlorides from carboxylic acids using $SOCl_2$

(f) use of acyl chlorides in synthesis in formation of esters, carboxylic acids and primary and secondary amides.

Including esterification of phenol, which is not readily esterified by carboxylic acids.